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Lecture 3 Linear Regression I

STAT 441: Statistical Methods for Learning and Data Mining

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Outline

1. Simple Linear Regression
2. Estimation and Inference
3. Summary and Remark

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Vectors

An array $\mathbf{x}$ of n real numbers $x_1, \ldots, x_n$ is called a *vector*, and it is written as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{or} \quad \mathbf{x}' = [x_1, x_2, \cdots, x_n].$$

sample size

- Here the prime denotes the operation of transposing a column to a row.
- The number n is referred to as the *dimension* of the vector $\mathbf{x}$.
- The *coordinates* $x_1, \ldots, x_n$ can also be complex numbers, in which case $\mathbf{x}$ is a complex vector. However in this subject we consider ONLY real numbers and real vectors.

- The set of all the real vectors of dimension n, is denoted by $\mathbb{R}^n$.
- One can scale a vector $\mathbf{x}$ by multiplying it by a constant c.

$$\mathbf{x} = x \quad \Rightarrow \quad c\mathbf{x} = cx.$$

- Vectors can be added.

$$\mathbf{x} + \mathbf{y} = x + y = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}.$$

- **inner product**:

$$\langle \mathbf{x}, \mathbf{y} \rangle := \sum_{i=1}^{n} x_i y_i = \langle \mathbf{y}, \mathbf{x} \rangle .$$

- **length**, or **norm**:

$$\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

- For any constant c,

$$\langle c\mathbf{x}, \mathbf{y} \rangle = c \langle \mathbf{x}, \mathbf{y} \rangle ;$$

$$\|c\mathbf{x}\| = \sqrt{\langle c\mathbf{x}, c\mathbf{x} \rangle} = \sqrt{c^2 \langle \mathbf{x}, \mathbf{x} \rangle} = |c| \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = |c| \|\mathbf{x}\|.$$

- **triangle inequality**: (proof left as a warming up exercise)

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

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Matrix

- **Matrix** A, of dimension $m \times n$, is just a table of (real) numbers.

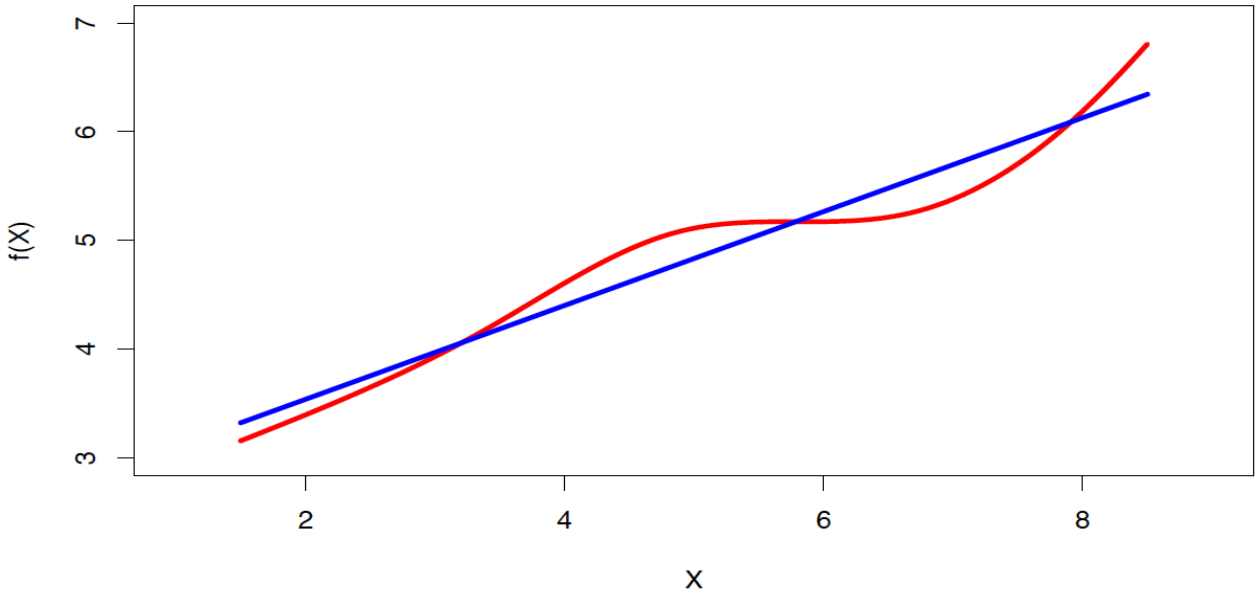
$$A = \begin{bmatrix} A_{1,1} & A_{1,2} & \cdots & A_{1,n} \\ A_{2,1} & A_{2,2} & \cdots & A_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & A_{m,2} & \cdots & A_{m,n} \end{bmatrix}.$$

- $A_{i,j}$ is called the (i,j)-entry of the matrix A.
- **Transpose**: A' is a $n \times m$ matrix, defined by

$$A' = \begin{bmatrix} A_{1,1} & A_{2,1} & \cdots & A_{m,1} \\ A_{1,2} & A_{2,2} & \cdots & A_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1,n} & A_{2,n} & \cdots & A_{m,n} \end{bmatrix}.$$

Simple Linear Regression

- Linear regression is a simple approach to supervised learning. It assumes that the dependence of Y on $X_1, X_2, \cdots, X_p$ is linear.
- True regression functions are never linear! although it may seem overly simplistic, linear regression is extremely useful both conceptually and practically.



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Simple Linear Regression

- Simple Linear Regression Model (SLR) has the form of

$$Y = \beta_0 + \beta_1 X + \varepsilon,$$

where β_0 and β_1 are two unknown parameters (coefficients), called intercept and *slope*, respectively, and ε is the error term.

- Given the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, the estimated regression line is

$$y = \hat{\beta}_0 + \hat{\beta}_1 x.$$

- For $X = x$, we predict Y by $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, where the hat symbol denotes an estimated value.

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Estimate the parameters

- Let (y_i, x_i) be the i-th observation and $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, we call $e_i = y_i - \hat{y}_i$ the ith residual.
- To estimate the parameters, we minimized the residual sums of squares (RSS),

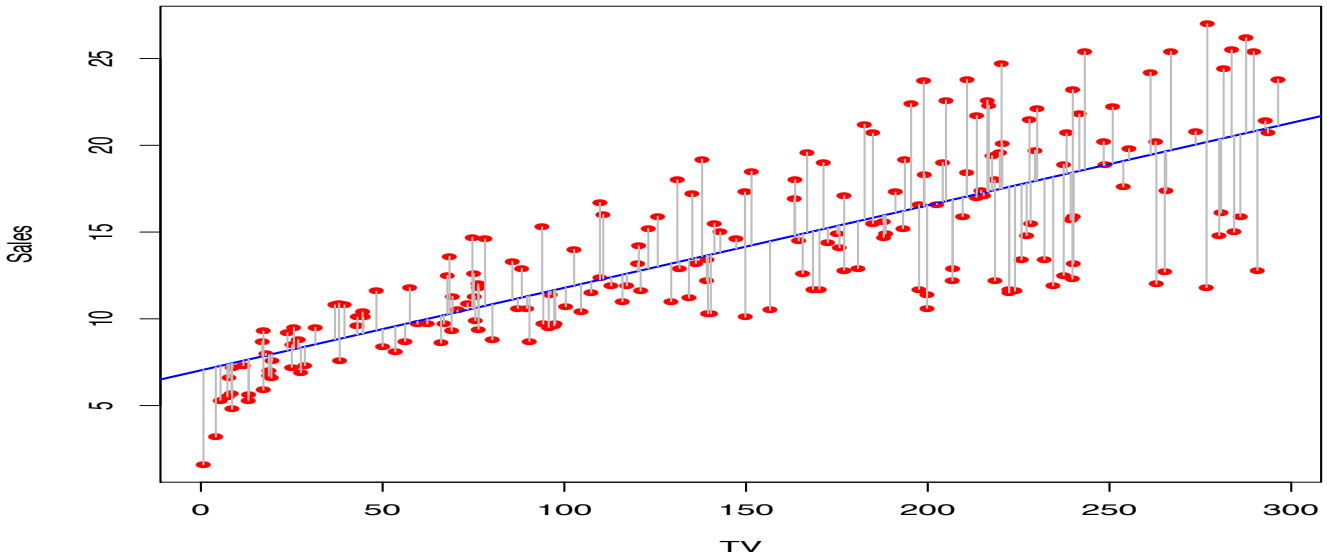
$$\text{RSS} = \sum_i e_i^2 = \sum_i \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i, \right)^2.$$

- Denote $\bar{y} = \sum_i y_i / n$ and $\bar{x} = \sum_i x_i / n$. The minimized values are

$$\hat{\beta}_1 = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} = \left(r \frac{\sqrt{\sum_i (y_i - \bar{y})^2}}{\sqrt{\sum_i (x_i - \bar{x})^2}} \right),$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Example



- Advertising data: the least square fit for the regression of `sales` and `TV`.
- Each grey line segment represents an error, and the fit makes a compromise by averaging their squares.
- In this case a linear fit captures the essence of the relationship, although it is somewhat deficient in the left of the plot.

Interval Estimates

A point estimate gives a single number, *but does not give information about the precision and reliability of estimation.* For example, given $\hat{\beta}_1 = 1.2231$, we have no information about how close it might be to β_1.

A 95% **confidence interval** (CI) for β_1 is an interval derived from sample, such that if we independently repeat this experiments many times, in 95% of our experiments the intervals obtained will cover β_1.
The percentage "95%" is called the **confidence level** or **confidence coefficient**. Usually people use 90%, 95%, 99%, etc, as confidence levels.

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Assess the coefficient estimates

- The standard error of an estimator reflects how it varies under repeated sampling.

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{\frac{\sigma^2}{\sum(x_i - \bar{x})^2}}, \qquad \mathrm{SE}(\hat{\beta}_0) = \sqrt{\sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right)},$$

  where $\sigma^2 = \mathrm{Var}(\varepsilon)$.
- A 95% confidence interval is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter.
- It has the form

$$\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1).$$

- For the advertising data, the 95% confidence interval for β_1 is $[0.042, 0.053]$, which means, there is approximately 95% chance this interval contains the true value of β_1 (under a scenario where we got repeated samples like the present sample).

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Hypothesis testing

- Definition: A **statistical hypothesis** is an assertion about the distribution of one or more random variables. If the statistical hypothesis completely specifies the distribution, it is called a **simple statistical hypothesis**; if it does not, it is called a **composite statistical hypothesis**.
- Definition: A **test** of statistical hypothesis is a decision rule which, when the experimental sample values have been obtained, leads to a decision to accept or to reject the hypothesis under consideration.

We do not expand all the details. For a comprehensive review, see any statistics text books, for example Morris H. DeGroot and Mark J. Schervish, *Probability and Statistics*, Pearson, 2012.

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Hypothesis testing

- Standard errors can also be used to perform hypothesis tests on the coefficients. The most common hypothesis test involves testing the null hypothesis of

  H_0: There is no relationship between X and Y versus the alternative hypothesis

  H_A: There is some relationship between X and Y.
- Mathematically, we test

$$H_0 : \ \beta_1 = 0 \text{ versus } H_A : \beta_1 \neq 0,$$

since if $\beta_1 = 0$ then the model reduces to $Y = \beta_0 + \varepsilon$, and X is not associated with Y.

Four possibilities

Decision Table for a Test of Hypothesis

	True State of Nature	
Decision	H_0 is true	H_1 is true
Reject H_0	Type I Error	Correct Decision
Accept H_0	Correct Decision	Type II Error

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Hypothesis testing

- To test the null hypothesis, we compute a t-statistics,

$$t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}\left(\hat{\beta}_1\right)}.$$

- This statistics follows t_{n-2} under the null hypothesis $\beta_1 = 0$.
- Using statistical software, it is easy to compute the probability of observing any value equal to $|t|$ or larger. We call this probability the p-value (the smallest level α_0 that we would reject the null-hypothesis (H_0) at level α_0 with the data.)

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.032594   0.457843   15.36   <2e-16 ***
TV          0.047537   0.002691   17.67   <2e-16 ***
---
Signif. codes:  0 ?***? 0.001 ?**? 0.01 ?*? 0.05 ?.? 0.1 ? ? 1
```

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Measure of fit

- We compute the Residual Standard Error

$$\mathrm{RSE} = \sqrt{\frac{1}{n-2}\mathrm{RSS}} = \sqrt{\frac{1}{n-2}\sum_i (y_i - \hat{y}_i)^2},$$

where the residual sum-of-squares is $\mathrm{RSS} = \sum_i (y_i - \hat{y}_i)^2$.

- R-squared or fraction of variance explained is

$$\mathrm{R}^2 = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}},$$

prediction is only included here.

where $\mathrm{TSS} = \sum_i (y_i - \bar{y})^2$ is the total sum of squares.

  - a statistical measure that represents the proportion of the variance for a dependent variable that's explained by an independent variable or variables in a regression model.

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Measure of fit

- It can be shown that in this simple linear regression setting that $\mathrm{R}^2 = r^2$, where r is the correlation between Y and X:

$$r = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_i (y_i - \bar{y})^2}\sqrt{\sum_i (x_i - \bar{x})^2}} = \left(\hat{\beta}_1 \frac{\sqrt{\sum_i (x_i - \bar{x})^2}}{\sqrt{\sum_i (y_i - \bar{y})^2}}\right).$$

Summary and Remark

- Simple linear regression
- Estimation and inference
- Measure of fit R^2
- Read textbook Chapter 3